

ANTIQUE PAST EXAMINATION QUESTIONS (FP1)

TOPIC 7: MATRICES (1)

1. The matrix M is given by $M = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}$, where a and b are positive constants.
- (a) The matrix M represents a sequence of two geometrical transformations.
- State the type of each transformation, and make clear the order in which they are applied. **(One way stretch followed by a shear.)** [2]
- The unit square in the x - y plane is transformed by M onto parallelogram $OPQR$.
- (b) Find, in terms of a and b , the matrix which transforms parallelogram $OPQR$ onto the unit square. **$\left(M^{-1} = \frac{1}{a} \begin{pmatrix} 1 & -b \\ 0 & a \end{pmatrix}\right)$** [2]
- It is given that the area of $OPQR$ is 2 cm^2 and that the line $x + 3y = 0$ is invariant under the transformation represented by M .
- (c) Find the values of a and b . **($a = 2, b = 3$)** [5]

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2. The matrices A and B are given by

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2}\sqrt{3} \\ \frac{1}{2}\sqrt{3} & \frac{1}{2} \end{pmatrix}.$$

- (a) Give full details of the geometrical transformation in the x - y plane represented by A . **(Reflection in the line $y = x$)** [1]
- (b) Give full details of the geometrical transformation in the x - y plane represented by B . **(Rotation of $\frac{1}{3}\pi$ anticlockwise about the origin.)** [2]

The triangle DEF in the x - y plane is transformed by AB onto triangle PQR .

- (c) Show that the triangles DEF and PQR have the same area. [3]
- (d) Find the matrix which transforms triangle PQR onto triangle DEF . **$\left((AB)^{-1} = \begin{pmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}\right)$** [2]

- (e) Find the equations of the invariant lines, through the origin, of the transformation in the x - y plane represented by AB . **($y = (2 - \sqrt{3})x, y = -(2 + \sqrt{3})x$)** [5]

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3. Let $A = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$.
- (a) The transformation in the x - y plane represented by A^{-1} transform a triangle of area 30 cm^2 into a triangle of area $d \text{ cm}^2$.

Find the value of d . **($\det A^{-1} = \frac{1}{2}; 15$)** [3]

- (b) Prove by mathematical induction that, for all positive integers n ,

$$A^n = \begin{pmatrix} 2^n & 0 \\ 2^n - 1 & 1 \end{pmatrix}. \quad [5]$$

- (c) The line $y = 2x$ is invariant under the transformation in the x - y plane represented by $A^n B$, where $B = \begin{pmatrix} 1 & 0 \\ 3 & 0 \end{pmatrix}$.

Find the value of n . ($n = 5$)

[5]

May/June 2020/11/12/6

4. The matrix A is given by

$$A = \begin{pmatrix} k & 0 & 2 \\ 0 & -1 & -1 \\ 1 & 1 & -k \end{pmatrix},$$

where k is a real constant.

- (a) Show that A is non-singular.

[3]

The matrices B and C are given by

$$B = \begin{pmatrix} 0 & -3 \\ -1 & 3 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad C = \begin{pmatrix} -3 & -1 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

It is given that $CAB = \begin{pmatrix} -2 & -\frac{3}{2} \\ -1 & -\frac{3}{2} \end{pmatrix}$.

- (b) Find the value of k . ($k = -\frac{1}{2}$) [3]

- (c) Find the equations of the invariant lines, through the origin, of the transformation in the x - y plane represented by CAB . ($y = -x, 3y - 2x = 0$) [5]

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5. The matrix A is given by

$$A = \begin{pmatrix} 5 & k \\ -3 & -4 \end{pmatrix}.$$

- (a) Find the value of k for which A is singular. ($\frac{20}{3}$) [2]

It is given that $k = 6$ so that $A = \begin{pmatrix} 5 & 6 \\ -3 & -4 \end{pmatrix}$.

- (b) Find the equations of the invariant lines, through the origin, of the transformation in the $x - y$ plane represented by A . ($y = -x, 2y + x = 0$) [6]

- (c) The triangle DEF in the $x - y$ plane is transformed by A onto triangle PQR .

- (i) Given that the area of triangle DEF is 10 cm^2 , find the area of triangle PQR . (20) [2]

- (ii) Find the matrix which transforms triangle PQR onto triangle DEF . [2]

$$\begin{pmatrix} 2 & 3 \\ -3/2 & -5/2 \end{pmatrix}$$

Specimen Paper 2020/1/5

6. Let A be the matrix $\begin{pmatrix} 17 & 12 \\ 12 & 10 \end{pmatrix}$.
- (a) (i) Determine the integer n for which $27A - A^2 = nI$, where I is the 2×2 identity matrix. (26) [2]
 (ii) **Hence** find A^{-1} in the form $pA + qI$ for rational numbers p and q . ($p = \frac{27}{26}, q = -\frac{1}{26}$) [2]
- (b) The plane transformation T is defined by $T: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto A \begin{pmatrix} x \\ y \end{pmatrix}$. It is given that T is a stretch, with scale factor k , parallel to the line $y = mx$, where $m > 0$.
 (i) Find the value of k . (26) [2]
 (ii) By considering $A \begin{pmatrix} x \\ mx \end{pmatrix}$, or otherwise, determine the value of m . ($\frac{3}{4}$) [4]

CAIE 9795 May/June 2018/1/11

7. (i) (a) Given that $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$, work out the matrix AB and write down expressions for $\det A$ and $\det B$. ($\det(A) = ad - bc$; $\det(B) = eh - fg$) [2]
 (b) Verify, by direct calculation, that $\det(AB) = \det(A) \times \det(B)$. [2]

CAIE 9795 May/June 2017/1/11

8. (i) Show that there is a value of t for which AB is an integer multiple of the 3×3 identity matrix I , where

$$A = \begin{pmatrix} 1 & 2 & 1 \\ t & 1 & -t \\ 3 & 2 & 1 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} t-2 & 0 & 5 \\ 12 & -2 & -6 \\ 3t & 4 & 7 \end{pmatrix} \quad (t = -3) \quad [4]$$

CAIE 9795 May/June 2014/1/2 (part)

9. The plane transformation T is the composition (in this order) of
- A reflection in the line $y = x \tan \frac{1}{8} \pi$; followed by
 - A shear parallel to the y -axis, mapping $(1,0)$ to $(1,2)$; followed by
 - A clockwise rotation through $\frac{1}{4} \pi$ radians about the origin; followed by
 - A shear parallel to the x -axis, mapping $(0,1)$ to $(-2,1)$.

Determine the matrix M which represents T , and hence give a full geometrical description of

T as single plane transformation. ($\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, reflection in $y = x$) [8]

CAIE 9795 May/June 2013/1/9

10. (i) Write down the 2×2 matrices which represent the following plane transformations:
 (a) an anticlockwise rotation about the origin through an angle α . ($\begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$) [2]
 (b) a reflection in the line $y = x \tan \left(\frac{1}{2} \beta\right)$. ($\begin{pmatrix} \cos \beta & \sin \beta \\ \sin \beta & -\cos \beta \end{pmatrix}$) [2]
- (ii) A reflection in the $x - y$ plane in the line $y = x \tan \left(\frac{1}{2} \theta\right)$ is followed by a reflection in the line $y = x \tan \left(\frac{1}{2} \phi\right)$. Show that the composition of these two reflections (in this order) is a rotation and describe this rotation fully. ($\begin{pmatrix} \cos(\phi - \theta) & -\sin(\phi - \theta) \\ \sin(\phi - \theta) & \cos(\phi - \theta) \end{pmatrix}$) [6]

$$-6-k$$

$$-6-(-7)=1$$

CAIE 9795 May/June 2012/1/5

MIA

11. Given that the matrix $A = \begin{pmatrix} 2 & k \\ 1 & -3 \end{pmatrix}$, where k is real, is such that $A^3 = I$, find the value of k and the numerical value of $\det A$. ($k = 7, \det A = 1$) [4]

CAIE 9795 May/June 2011/1/1

12. The matrix C is given by $C = AB$, where

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}.$$

- (i) Evaluate C . $\begin{pmatrix} 4 & 3 \\ 3 & 4 \end{pmatrix}$ [1]
 (ii) Find the point of the x - y plane which, under the transformation represented by C , maps onto the point with coordinates $(50, 100)$. $\begin{pmatrix} -4 \\ 22 \end{pmatrix}$ [3]

FM June 1996/1/2

13. Given that

$$A = \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 5 & -1 \\ 2 & -4 \end{pmatrix},$$

Find

- (i) $\det(AB)$, (144) [2]
 (ii) $\det(B^{-1})$. $\left(-\frac{1}{18}\right)$ [2]

Nov 1988/1/12

14. The matrix A is given by

$$A = \begin{pmatrix} 5 & 6 \\ -3 & -4 \end{pmatrix}.$$

Find the equation of the invariant lines (through an origin) of the transformation T given by

$$T: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto A \begin{pmatrix} x \\ y \end{pmatrix}. \quad [6]$$

$$(x + y = 0, x + 2y = 0)$$

Nov 1987/1/19 (part)

15. Under the transformation

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} 3 & 2 \\ -1 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

the non-linear points A, B, C are transformed to P, Q, R respectively. Find the value

$$\frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle PQR} \quad [3]$$

$$\left(\frac{1}{14}\right)$$

Nov 1987/1/13

16. (i) Write down the matrix A representing an anticlockwise rotation about the origin, through an angle α , of points in the $x - y$ plane (represented by column vectors). Describe the transformation represented by A^T . [3]

$$A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix};$$

A^T represents rotation about the origin, through α clockwise

- (ii) Write down the matrix B representing a reflection in the x -axis. Denoting ABA^T by C , show that

$$C = \begin{pmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{pmatrix}. \quad [3]$$

$$\left(B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right)$$

- (iii) In either order,
 (a) describe the transformation represented by C , (**reflection in the line $y = (\tan \alpha)x$**) [3]
 (b) determine the invariant lines of the transformation represented by C . (**$y = (\tan \alpha)x$, $y = -(\cot \alpha)x$**) [3]

Nov 1986/1/17

17. The matrix $\begin{pmatrix} 1 & 3 \\ 2 & -4 \end{pmatrix}$ is denoted by A . Find the invariant lines of the transformation T , where

$$T: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto A \begin{pmatrix} x \\ y \end{pmatrix}. \quad [4]$$

$$\left(y = \frac{1}{3}x, y = -2x \right)$$

June 1986/1/16 (part)

MA

18. Given that the matrix A is singular, where $A = \begin{pmatrix} 1 & 2 & 1 \\ a & -1 & -11 \\ -2 & a & 12 \end{pmatrix}$, find the possible values of a . **$a = 3, 10$**

FM June 1985/1/14 (part)

19. The matrices A and B are given by

$$A = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2}\sqrt{3} \\ \frac{1}{2}\sqrt{3} & \frac{1}{2} \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

- (i) The two transformation T_1 and T_2 of points of the $x - y$ plane are defined by

$$T_1: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto A \begin{pmatrix} x \\ y \end{pmatrix}, \quad T_2: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto B \begin{pmatrix} x \\ y \end{pmatrix},$$

Describe T_1 and T_2 geometrically. (**T_1 is an anticlockwise rotation of 60° about O . T_2 is a reflection in the x -axis.**)

- (ii) Hence, or otherwise, find the matrices A^6 and ABA^{-1} . $\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \right)$
- (iii) Given that $C = ABA^{-1}$, show that there are two lines through the origin which are invariant under the transformation T_3 , defined by

$$T_3: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto C \begin{pmatrix} x \\ y \end{pmatrix},$$

and give the equation of these lines. $(y = \sqrt{3}x, y = -\frac{1}{\sqrt{3}}x)$

Nov 1983/2/4

20. (a) The matrix given by $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where $a + d = -1$ and $ad - bc = 1$. Show that $M^2 = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$, and deduce that $M^2 + M + I = O$.
- (b) A transformation is given by $X \rightarrow AX$, where A is a 2×2 matrix and $X = \begin{pmatrix} x \\ y \end{pmatrix}$. The points $(1,0)$ and $(0,1)$ are mapped to $(1,0)$ and $(p, 1)$ respectively. Write down the matrix A .

A second transformation is given by $X \rightarrow BX$ and is an anticlockwise rotation about the origin through an obtuse angle θ , where $\cos \theta = -\frac{4}{5}$. Write down the matrix B .

Given that the matrix M in (a) above is such that $M = BA$, find the value of p .

FM Nov 1982/2/3

21. (b) Find the matrices A and B such that
- (i) the transformation $\begin{pmatrix} x \\ y \end{pmatrix} \mapsto A \begin{pmatrix} x \\ y \end{pmatrix}$ gives an anti-clockwise rotation about the origin through an angle α . $\begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$
- (ii) the transformation $\begin{pmatrix} x \\ y \end{pmatrix} \mapsto B \begin{pmatrix} x \\ y \end{pmatrix}$ gives a reflection in the line $y = x \tan \beta$. $\begin{pmatrix} \cos 2\beta & \sin 2\beta \\ \sin 2\beta & -\cos 2\beta \end{pmatrix}$

Obtain the matrix AB , expressing the elements in a simplified form, and describe the transformation given by this matrix. $\left(AB = \begin{pmatrix} \cos(\alpha + 2\beta) & \sin(\alpha + 2\beta) \\ \sin(\alpha + 2\beta) & -\cos(\alpha + 2\beta) \end{pmatrix} \right)$

Nov 1978/1/5 (part)

22. When column vectors, representing points in a plane with the usual axes, are premultiplied by the matrix

$$M = \begin{pmatrix} 3 & -2 \\ 6 & k \end{pmatrix},$$

a transformation of the plane results,

- (i) If $k = 3$, find
- (a) the point whose image is $(8,9)$, $(2, -1)$

- (b) the image of the line $y = x$. ($y = 9x$)
 (ii) If $k = -4$, find
 (a) the set of points which map onto the origin, $\{x, \frac{3}{2}x\}$
 (b) the image of the line $y = \frac{3}{2}x - \frac{1}{2}$. ($1, 2$)

June 1978/2/5

MA

23. (a) Solve, for x , the equation $\begin{vmatrix} 1 & x & x^3 \\ 1 & a & a^3 \\ 1 & b & b^3 \end{vmatrix} = 0, a \neq b$.

FM June 1875/2/1

MA

24. (a) Solve for λ $\begin{vmatrix} 3-\lambda & -5 & -1 \\ 2 & 2-\lambda & 0 \\ 2 & 5 & 1 \end{vmatrix} = 0$.

FM June 1974/2/5 (part)

25. (a) Prove that the determinant

$$\begin{vmatrix} a & a+b & a+2b \\ a+b & a+2b & a \\ a+2b & a & a-b \end{vmatrix},$$

where $b \neq 0$, is equal to zero if and only if $a = -b$.

- (b) Show that



$$\begin{vmatrix} b_1 + c_1 & c_1 + a_1 & a_1 + b_1 \\ b_2 + c_2 & c_2 + a_2 & a_2 + b_2 \\ b_3 + c_3 & c_3 + a_3 & a_3 + b_3 \end{vmatrix} = 2 \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}.$$

FM Nov 1973/2/5

MA

26. Find the inverse of the matrix A , where $A = \begin{pmatrix} 1 & -3 & -1 \\ 5 & -1 & 5 \\ -3 & 1 & 1 \end{pmatrix}$.

FM Nov 1973/2/20

MA

27. Find the inverse of the matrix A , where $A = \begin{pmatrix} 2 & 1 & -1 \\ 6 & -1 & -9 \\ 4 & 3 & 1 \end{pmatrix}$.

FM June 1972/2/20